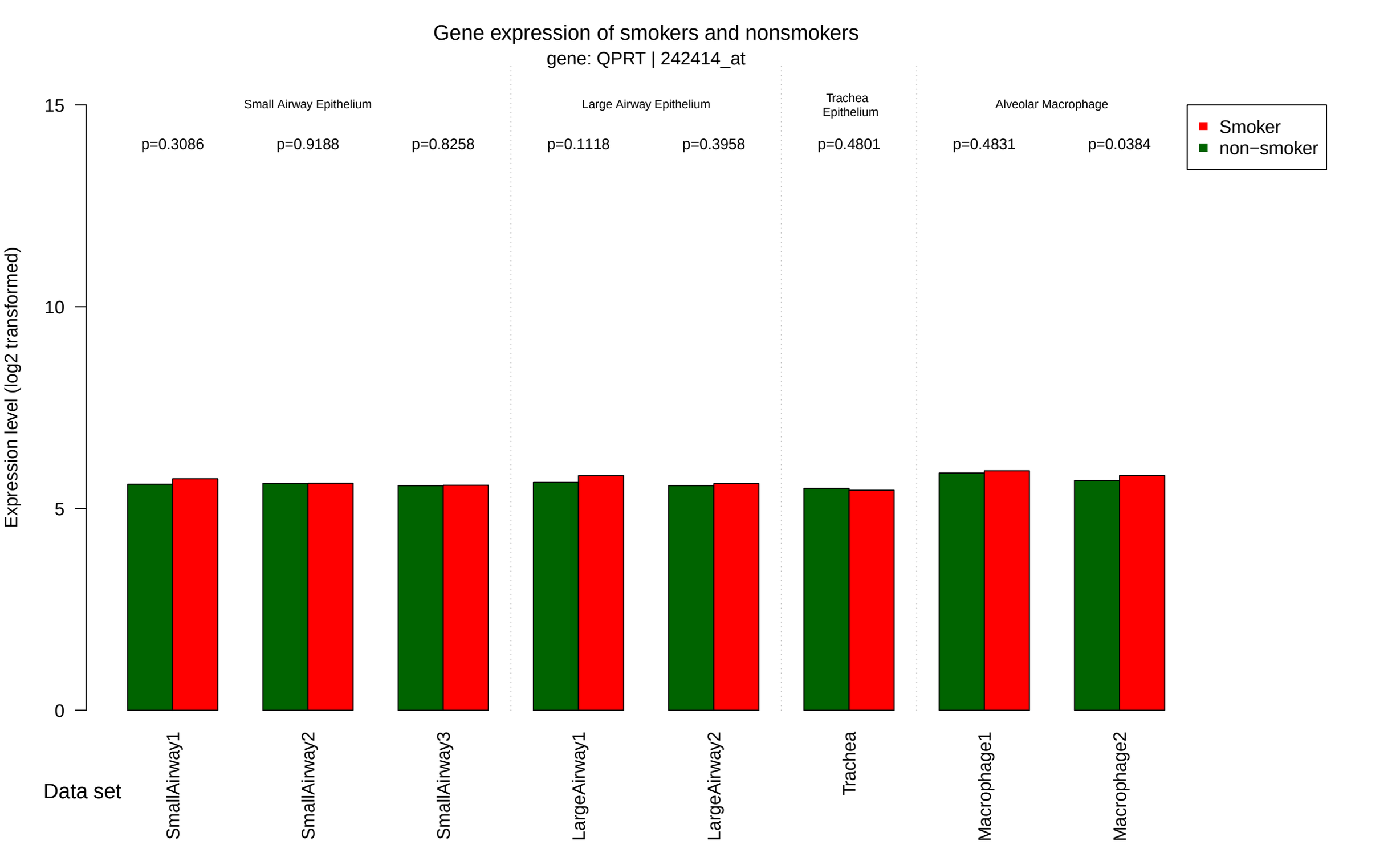
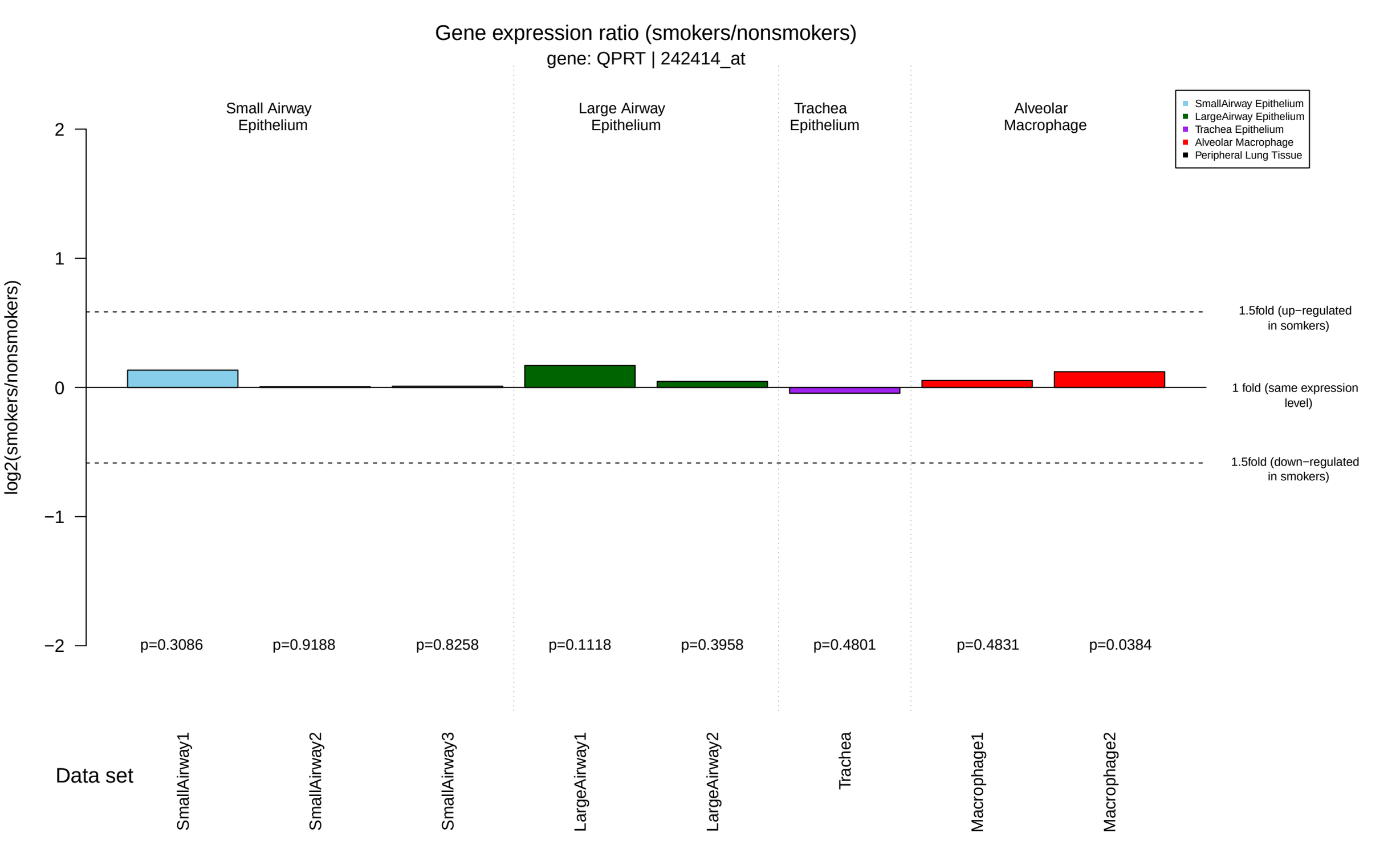
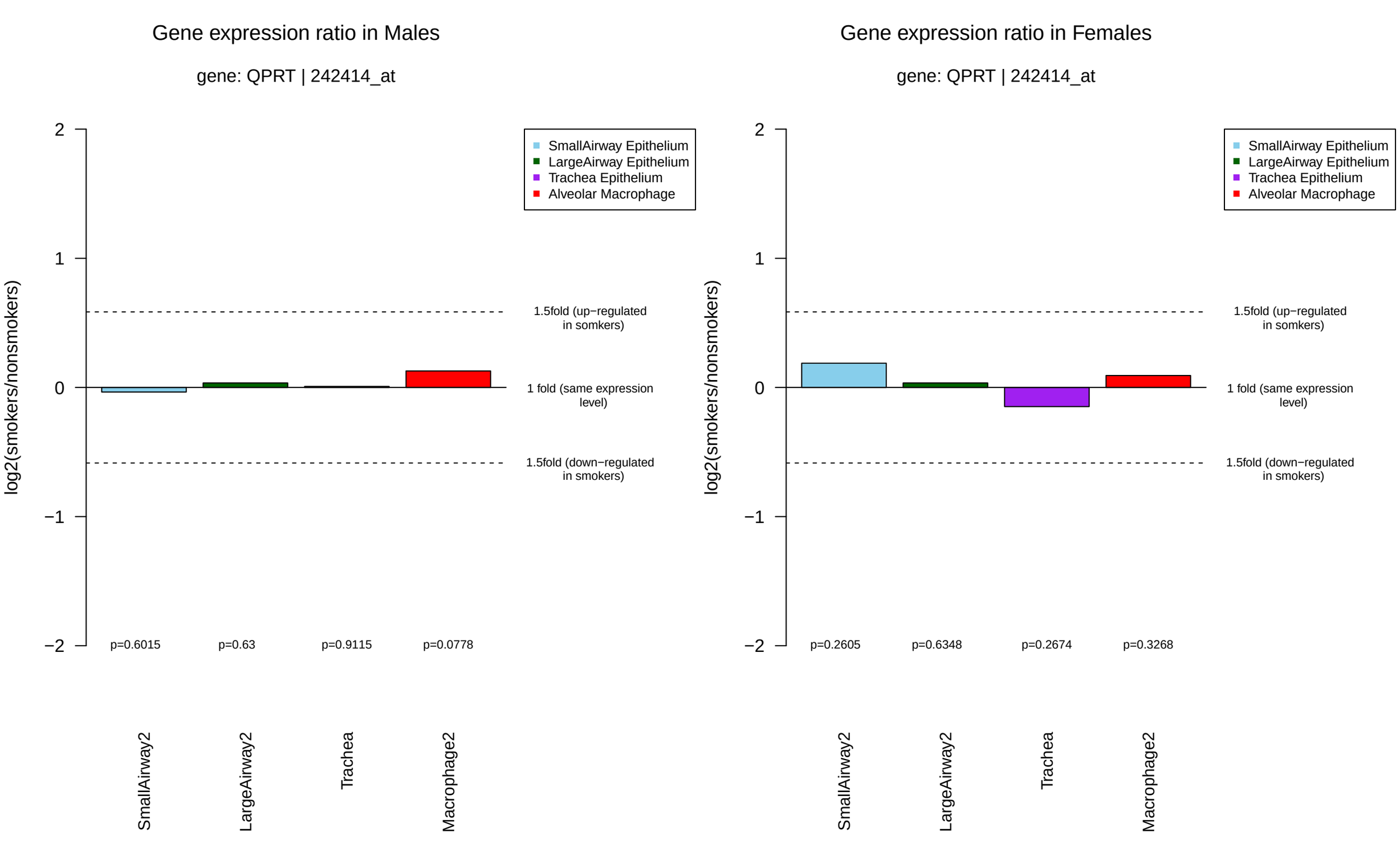
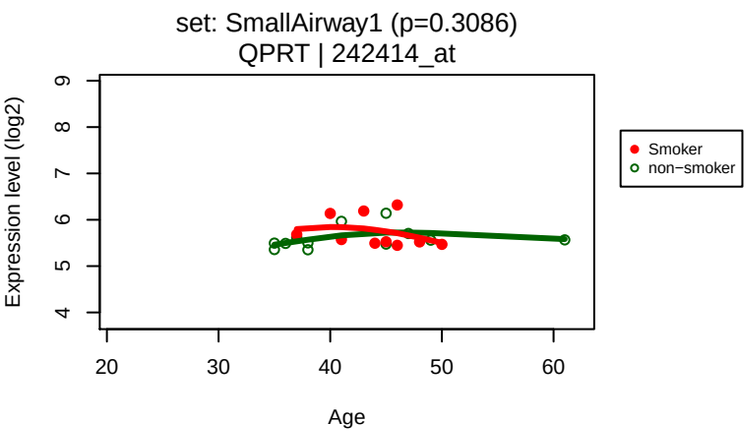
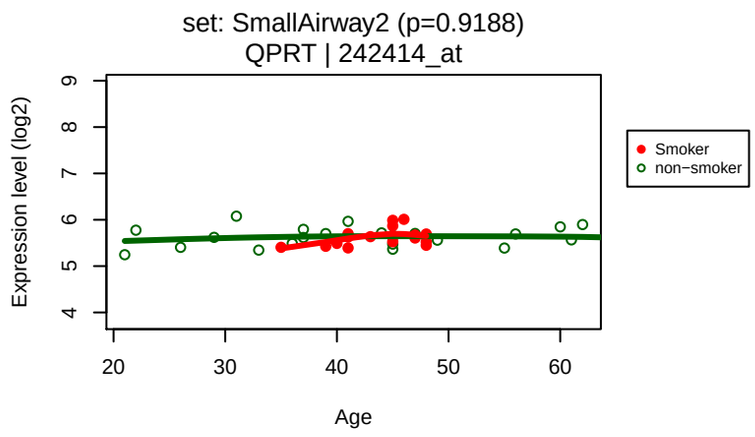
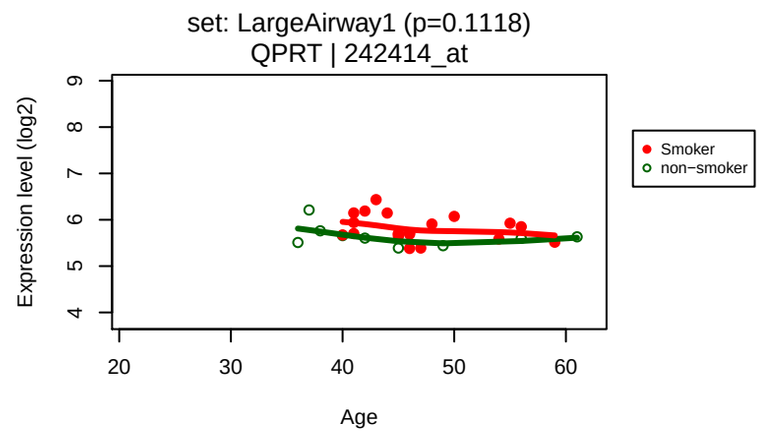

Gene expression of smokers and nonsmokers
gene: QPRT | 242414_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3086
p=0.9188
p=0.8258
p=0.1118
p=0.3958
p=0.4801
p=0.4831
p=0.0384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

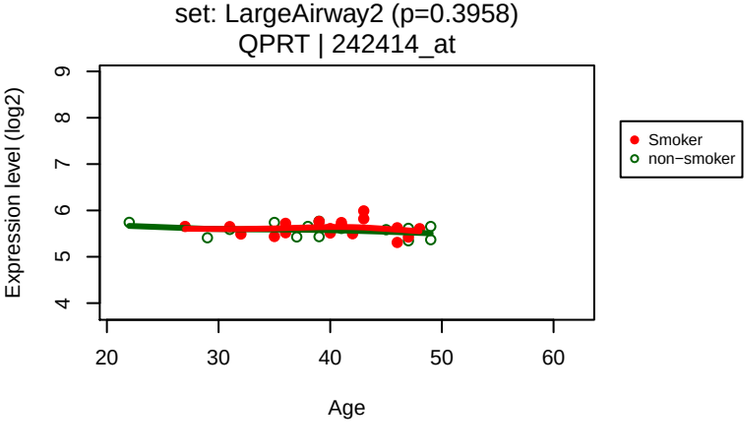

Gene expression ratio (smokers/nonsmokers)
gene: QPRT | 242414_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3086
p=0.9188
p=0.8258
p=0.1118
p=0.3958
p=0.4801
p=0.4831
p=0.0384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

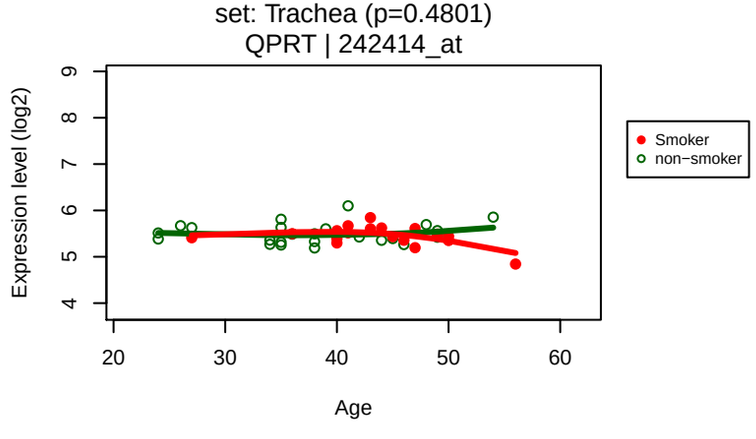

Gene expression ratio in Males
gene: QPRT | 242414_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6015
p=0.63
p=0.9115
p=0.0778
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: QPRT | 242414_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2605
p=0.6348
p=0.2674
p=0.3268
SmallAirway2
LargeAirway2
Trachea
Macrophage2

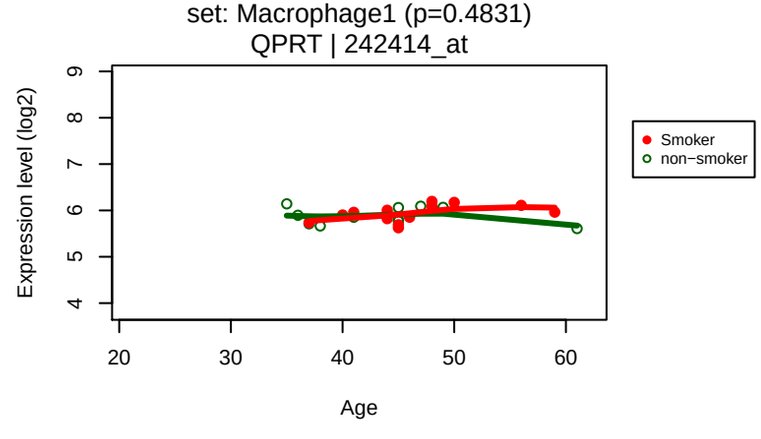

set: SmallAirway1 (p=0.3086)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker

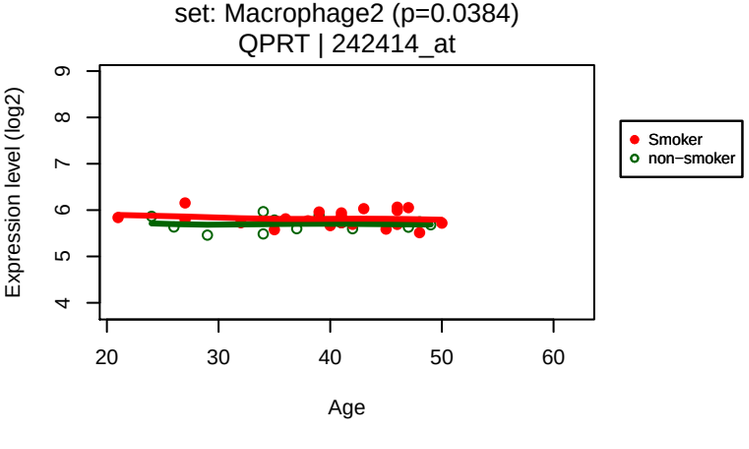

set: SmallAirway2 (p=0.9188)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker

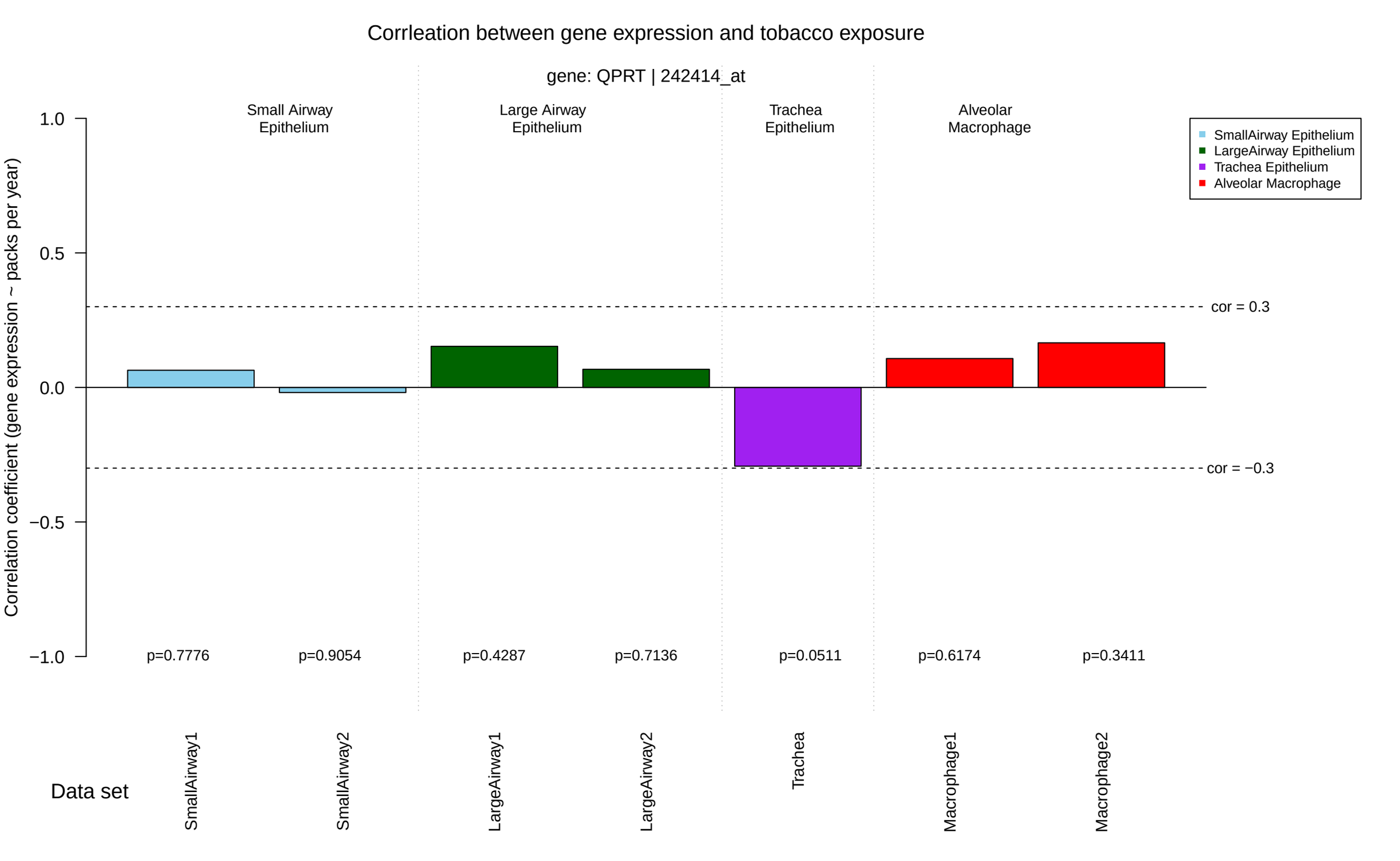

set: LargeAirway1 (p=0.1118)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3958)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4801)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4831)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0384)
QPRT | 242414_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: QPRT | 242414_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7776
p=0.9054
p=0.4287
p=0.7136
p=0.0511
p=0.6174
p=0.3411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2